

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1.5	1	10
Q2	2	3	2	1	1	9
Q3	2	3	2	1	1	9
Q4	2	3	2	1	1	9

Faculty Signature

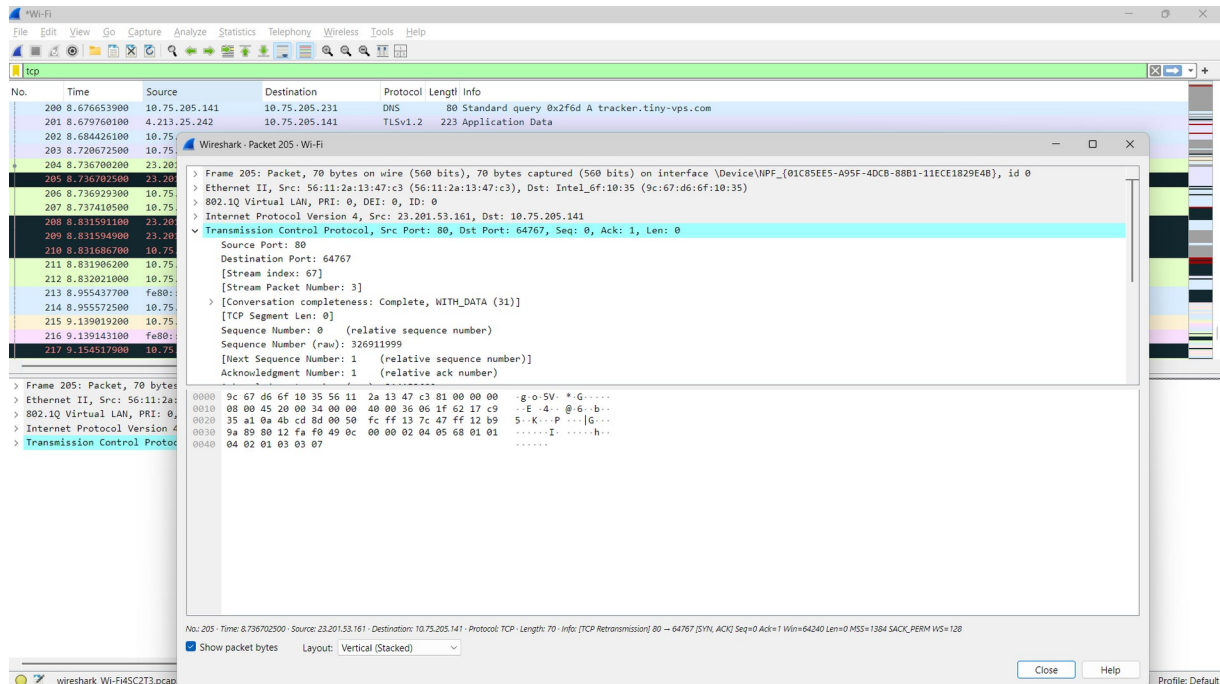
Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)

Screenshot – Question 1:

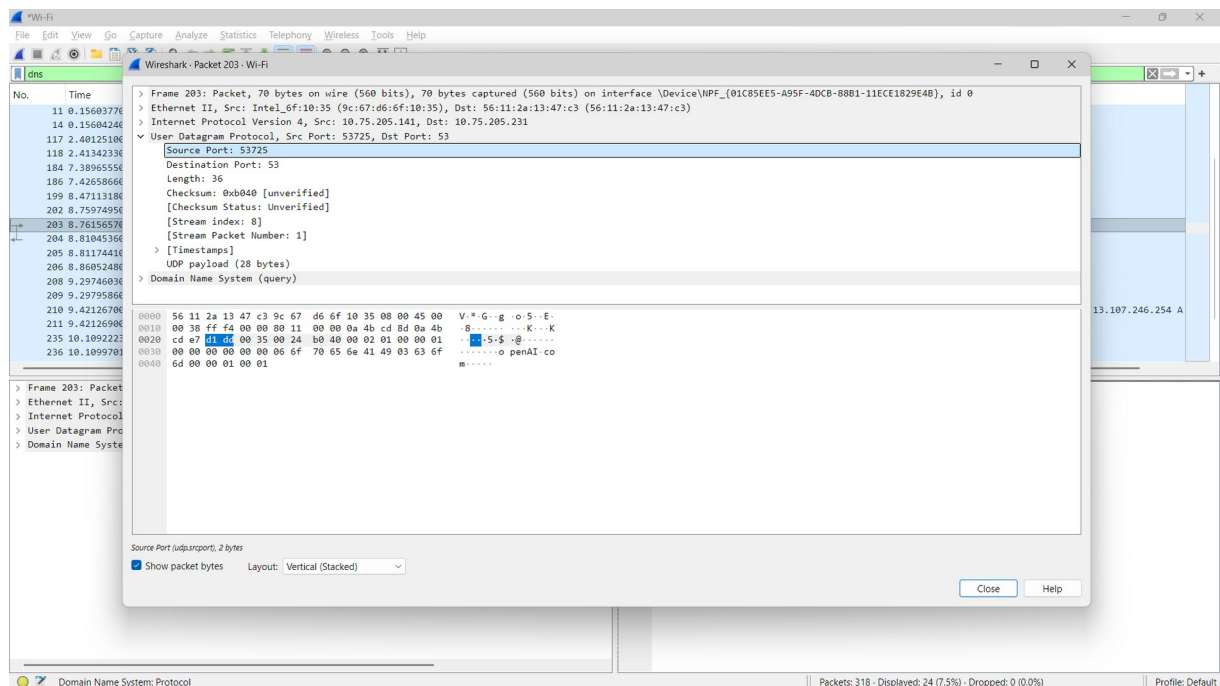


Wireshark capture of packet 205 — a retransmitted SYN, ACK segment (Src Port 80 → Dst Port 64767, Seq=0, Ack=1), showing the SYN and ACK flags used during TCP connection establishment.

The detail pane confirms this is part of the three-way handshake: the Info bar reads “[TCP Retransmission] 80 → 64767 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128”. The SYN flag signals a connection request/acknowledgment of synchronization, and the ACK flag (Acknowledgment Number: 1) confirms receipt of the client's initial SYN — together forming the second step of the handshake before the client's final ACK completes the reliable, connection-oriented setup.

2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)

Screenshot – Question 2:

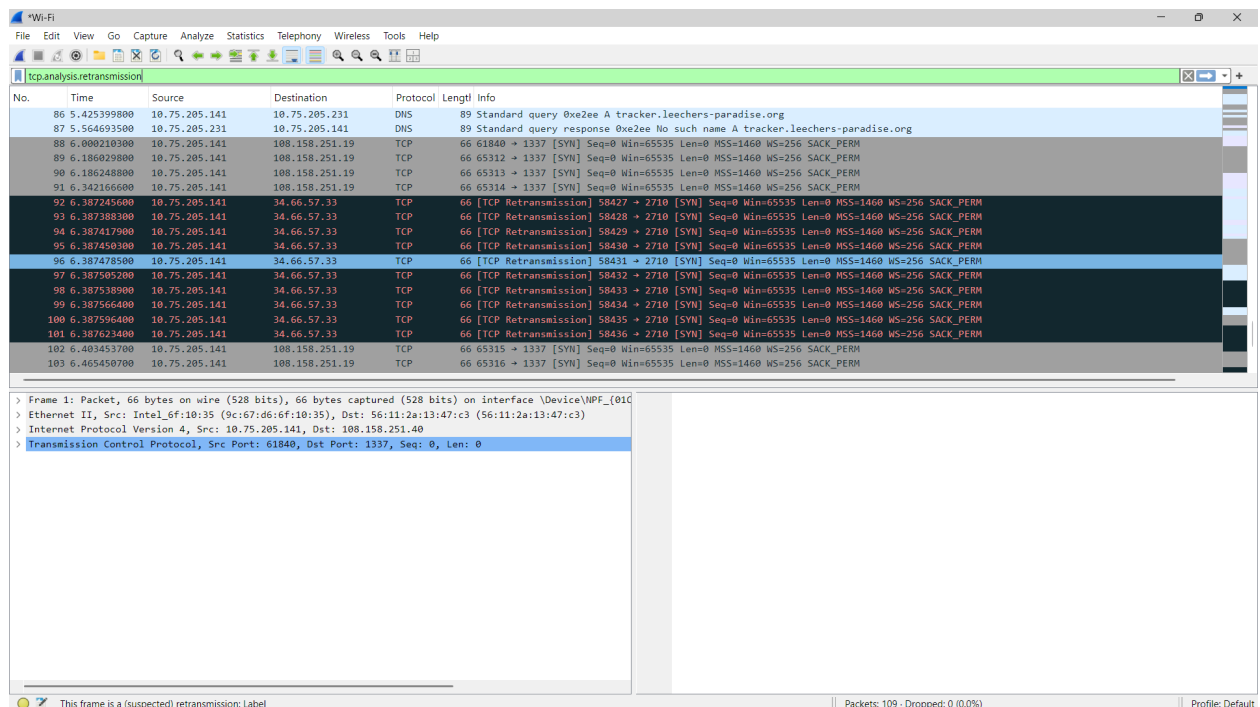


Wireshark capture of packet 203 — a DNS query carried over UDP (Src Port 53725 → Dst Port 53), with the UDP header fully expanded.

The UDP header shown here is only 8 bytes (Source Port, Destination Port, Length, Checksum) — visibly smaller than a TCP header's minimum 20 bytes — and, critically, it has no Sequence Number or Acknowledgment Number fields at all, unlike the TCP segment in Question 1. This lightweight, connectionless design is exactly why DNS uses UDP: a single query/response exchange needs minimal overhead and low latency, and does not require TCP's handshake or ordered-delivery guarantees for a one-shot lookup.

3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)

Screenshot – Question 3:

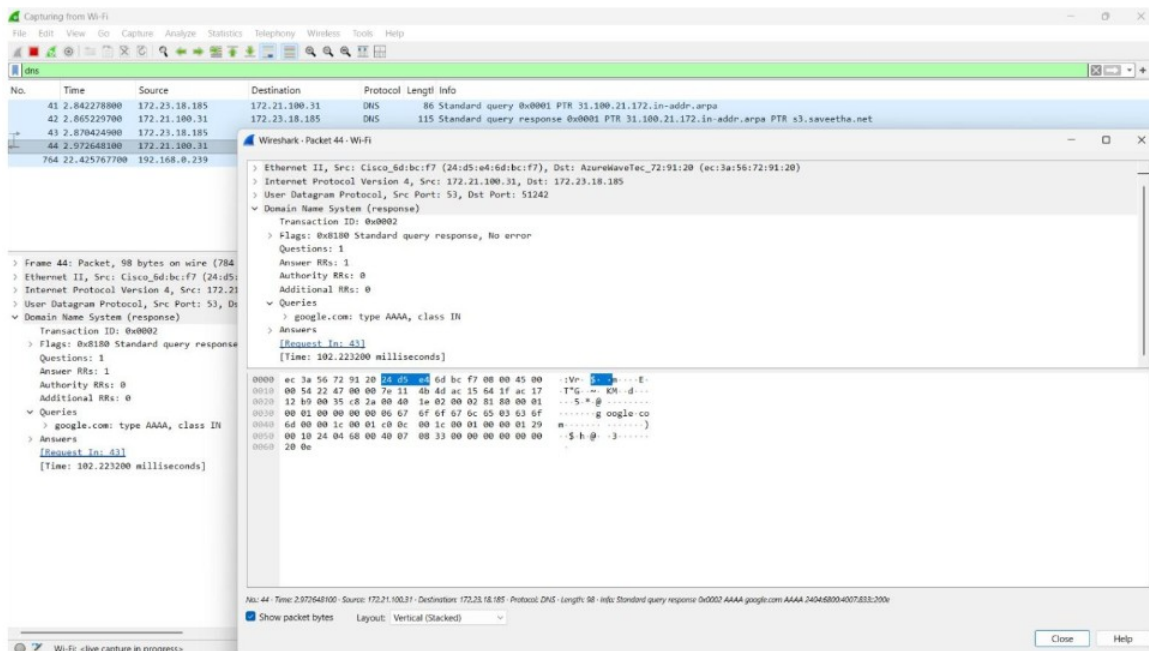


Wireshark filtered on `tcp.analysis.retransmission`, showing repeated [TCP Retransmission] SYN packets after simulated packet loss, alongside DNS queries that simply failed with “No such name”.

This capture demonstrates the core reliability difference directly: TCP (frames 92–101) retransmits the same SYN segment (58427–58436 → port 2710) repeatedly at increasing intervals after no response was received, actively working to recover the lost connection attempt. By contrast, the DNS query over UDP (frame 86, tracker.leechers-paradise.org) received a single definitive “No such name” response with no retransmission logic at all — UDP simply delivers or drops, leaving any recovery entirely up to the application. This is precisely why TCP is reliable but slower under loss (it spends time retrying) while UDP is faster but offers no delivery guarantee.

4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP. (BL4) (10)

Screenshot – Question 4:



Wireshark capture of packet 44 — a DNS response for google.com, showing an AAAA record answer and the measured resolution time.

The Queries section shows the query type is AAAA (an IPv6 address record) for google.com, and the Answers section confirms a successful response with no error. The detail pane's [Time: 102.223200 milliseconds] field (referencing the matching query in frame 43) directly measures the DNS resolution time for this lookup — about 102 ms. As in Questions 2 and 3, this query/response exchange is carried over UDP (Src/Dst Port 53), confirming why DNS typically prefers UDP: the exchange is a single small request/response pair, so UDP's lower overhead and lack of connection-setup delay directly translate into a faster resolution time than a TCP-based exchange would provide for the same lookup.